Registration No.:_____

Indian Institute of Technology Bhubaneswar
Autumn Mid-Semester Examination 2018

Subject: Signals and Systems (EC2L002)
BTech Programme: (EE and ECE)

Full Marks: 60
Date: 01-10-18 and Duration: 02 Hours

Answers All Questions (Assume parameters if required)

PART-A (10 x 2 = 20 Marks)

1. Determine if the following CT signals are periodic. If yes, compute the fundamental period T_0 and fundamental frequency for the CT signals:

   i) $x_1(t) = \sin(-5\pi t/8 + \pi/2)$ ii) $x_2(t) = \sin(6\pi t/7) + 2\cos(3\pi t/5)$

2. Determine if the following CT and DT signals are energy or power signals or neither.

   i) $x[n] = u[n] - u[n-1]$ ii) $x(t) = e^{j\omega t}$

3. How do you check the causality and stability of a LTI system using an impulse response, $h[n]$?

4. Evaluate the following integrals

   i) $\int_{-\infty}^{\infty}[\sin\left(\frac{3\pi t}{4}\right) + \exp(-4t+1)]\delta(-t-1)dt$

   ii) $\int_{-\infty}^{\infty}[u(t-4) - u(t-8)]\sin(3\pi t/4)\delta(t-5)dt$

5. Show that functions $cos(2\pi t)$ and $cos(3\pi t)$ are orthogonal over interval t=[0, 1].

6. Determine the response of the system

$$y[n] = \sum_{k=-\infty}^{n} x[k]$$

to the input sequence

$$x[n] = \begin{cases} |n|, & -3 \le n \le 3 \\ 0, & \text{otherwise} \end{cases}$$

7. Given the continuous-time signal specified by

$$x(t) = \begin{cases} 1 - |t|, & -1 \le t \le 1 \\ 0, & \text{otherwise} \end{cases}$$

Determine the resultant discrete-time sequence $x[nT_s]$ obtained by uniform sampling of $x(t)$ with a sampling period T_s of 0.25 s. Sketch the discrete sequence $x[n]$ for the sample index, $-5 < n < 5$.

8. Find and sketch the first derivatives of the following signals
   (a) $x(t) = u(t) - u(t-a), \quad a > 0$
   (b) $x(t) = t[u(t) - u(t-a)], \quad a > 0$

9. Sketch the waveform $y(t) = x(2t-6)$ for $x(t)$ as shown in Fig.1.

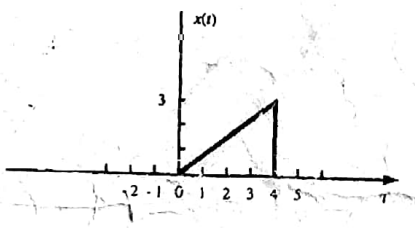

Figure 1

10. What is deterministic and random signal? Give suitable examples.

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PART-B (3 x 10 = 30 Marks)

11. **(A)** Determine the impulse response for the cascade of two linear time-invariant systems having the impulse responses $h_1[n] = \left(\frac{1}{2}\right)^n u[n]$ and $h_2[n] = \left(\frac{1}{4}\right)^n u[n]$.

(B) Consider the input signal $x[n]$ and the impulse response $h[n]$ shown in Fig.2.

Figure 2.

Compute the output $y[n]$.

12. Consider the LTI continuous system described by the differential equation:

$$\frac{d^2y(t)}{dt^2} + 4\frac{dy(t)}{dt} + 3y(t) = \frac{dx(t)}{dt} + 2x(t)$$

a. By using CTFT, find the impulse response of the system, $h(t)$.
b. If the input signal is $x(t) = e^{-t}u(t)$, what should be the output $y(t)$ of the LITC system?

13. **(A)** Compute the autocorrelation of the discrete-time signal

$$x[n] = a^n u[n], \quad 0 < a < 1$$

and sketch the normalized autocorrelation sequence for the lag values, $-5 < l < 5$.

(B) Write the properties of autocorrelation function.

PART-C (10 x 1 = 10 Marks)

14. The Smartphone system consists of the following sensors: _____, _____, _____ and _____.

15. The wireless channel is an example for_____(time-variant/time-invariant) system for longer observation interval.

16. The pitch frequency range of human voice is _____. The voice signal is recorded using_____ sensor.

17. The _____ and _____ are the practical examples for the 1-D and 2-D signals, respectively.

18. What is Gibbs phenomenon?

19. The biosignals such as _____ and _____ are recorded for diagnosis of the cardiovascular diseases.

20. How do you determine whether the system is invertible or non-invertible?

21. Define incrementally linear system with suitable example?

22. The magnitude responses of original and time-delayed signals are same. (True/False)

23. In a signal processing front-end module, _____ process converts the continuous-time signal into the discrete-time signal and _____ process converts the analog amplitude values into the discrete amplitude values.

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Indian Institute of Technology Bhubaneswar

School of Minerals, Metallurgical and Materials Engineering

Mid Semester (Autumn) Examination – 2018

Subject Name: Introduction to Materials Science and Engineering **Subject Code:** ID2L001

Date: 24-September-2018

Duration: 2 Hours **Full Marks:** 75

Instructions:

- All questions are to be answered. Answer a new question on a fresh page (strictly) and all parts of a question together. Draw all the figures as clearly as possible.
- Write all the steps clearly and give explanation for complete credit

List of constants:

- *Boltzmann's constant, $k_B = 1.38 \times 10^{-23}$ J/atom-K*
- *Gas constant, R = 8.314 J/mol-K*

I. Choose the correct answer(s). *There can be more than one correct answer.* [1×15=15]

1. Potential energy of a metal for $r < r_0$(equilibrium interatomic distance) is dominated by ____ between atoms
   (A) repulsion (B) attraction (C) difference in electronegativity (D) None
2. The valance electrons in metallic bonding are
   (A) shared among neighbouring kernels (B) delocalized
   (C) transferred to the neighbours (D) remain with the atom
3. For a FCC unit cell, the ratio of the number of tetrahedral voids to the number of atoms is:
   (A) 2:1 (B) 3:1 (C) 4:1 (D) 5:1
4. Indium has FCC structure. Its density and atomic weight are 22400 kg/m^3 and 192.2, respectively. The atomic radius of iridium in nm is
   (A) 0.126 (B) 0.136 (C) 0.146 (D) 0.156
5. Polymorphism means _____
6. Crystal structure of gold is FCC with a lattice parameter of 4.08 Å. The spacing between two closely spaced (123)-type planes is _____
7. Defect(s) that is/are bounded by two mirror planes is/are
   (A) twin (B) stacking fault (C) grain boundary (D) edge dislocation
8. The angle between lattice vectors [110] and [111] in degrees is _____.
9. The planar density of atoms on a (110) plane of a FCC crystal is:
   (A) 0.68 (B) 0.74 (C) 0.85 (D) 0.91
10. For an edge dislocation, the region below the extra half plane is under
    (A) compression (B) tension (C) no stress (D) None
11. A phase is _____
12. Gibb's phase rule is
    (A) P+C=F+N (B) F-N=C-P (C) P+F=C-N (D) P+N=C+F

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13. The dislocation line is parallel to the Burgers vector in _____ dislocation.

14. The fraction of vacancies in equilibrium close to the melting point of an Aluminium crystal (melting point = 660 °C and Q = 68 kJ/mole) is

(A) 1.55×10^{-4} (B) 1.56×10^{-5} (C) 1.58×10^{-4} (D) none of these

15. The reaction that results in two new solid phases from a solid upon cooling is

(A) Eutectic (B) Eutectoid (C) Peritectic (D) Peritectoid

II. Answer the following questions clearly and briefly to the point.

a) Calculate the diameter of the largest sphere that fits at the centre of a cube edge of a BCC unit cell with lattice parameter *a*. [2]

b) Is there any plane in a FCC lattice that has similar atomic packing as (0001) of a HCP crystal? [2]

c) What are octahedral and tetrahedral voids? [2]

d) If the vacancy concentration in a crystal triples itself between 27 °C and 47 °C, what is the enthalpy of formation (also known as the activation energy) of vacancies in kJ/mole?

e) Is it possible for two edge dislocations of opposite sign to annihilate each other? Explain your answer. [2]

f) For each of edge, screw, and mixed dislocations, cite the relationship between the direction of the applied shear stress and the direction of dislocation line motion. [2]

g) Consider two edge dislocations of opposite sign. Imagine that their slip planes do not coincide but are separated by one interatomic distance. Imagine that the dislocations are moved towards each other. Draw and briefly describe the defect resulting when the two edge dislocations are aligned vertically. [3]

h) What kind of a 2D defect retains a crystallographic relationship between the neighbouring crystal regions? Explain briefly. [2]

i) Which among the four elements, H, Ni, C, and W, can be the fastest diffusing species in Fe? Justify your answer. [1]

j) What happens if a single phase, multi-grained (polycrystalline) metallic sample is heated to an elevated temperature but within the solid state, and kept for long time? Why? [3]

k) What is coring? How to get rid of a cored microstructure, and why would you like to get rid of it? [3]

l) Aluminium has a face-centred cubic structure. The lattice parameter of aluminium is 4.05 Angstrom, and atomic weight of aluminium is 26.98 g/mole. Then, calculate:

1. the atomic radius of aluminium. [1]
2. the effective number of aluminium atoms in a unit cell. [1]
3. Calculate the theoretical density of aluminium in kg/m^3. [2]
4. Calculate the atomic packing factor of a unit cell of aluminium. [1]

m) Calculate the packing efficiency/factor of HCP unit cell. [3]

n) For a K^+ - Cl^- ion pair, the attractive (E_A) and repulsive (E_R) energies depend on the distance between the ions:

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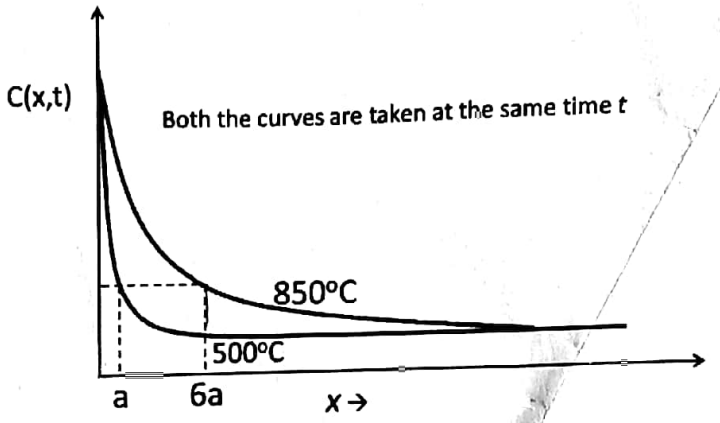
$$E_A = -\frac{1.436}{r}$$

$$E_R = \frac{5.86 \times 10^{-6}}{r^9}$$

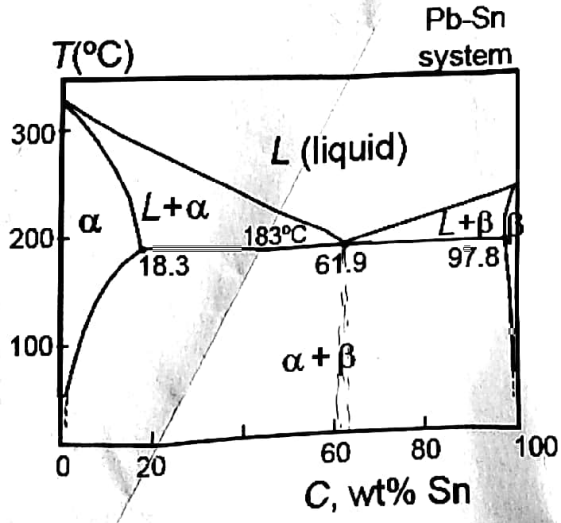
Both attractive and repulsive energies have units of eV [1 eV = $1.60217657 \times 10^{-19}$ Joules] and the distance is expressed in nm. Determine (i) the equilibrium spacing between K^+ and Cl^- ions, and (ii) magnitude of the bond energy E_0 between the two ions. [5]

o) The ratio of diffusion rate of silver in silicon at 1450 °C to that at 1000 °C was found to be 10 in a doping process. Calculate the activation energy Q for silver diffusion in silicon. [5]

p) In an experiment performed at different temperatures, the composition profiles along the depth at a given instant of time are as shown in the picture below. Find the activation energy for diffusion. [5]



q) In the Pb-Sn phase diagram shown below, what is the volume fraction of Eutectic mixture and proeutectic α just below the eutectic temperature? [3]



r) In three-component alloy, what is the maximum number of phases that can co-exist under equilibrium at any temperature? [3]

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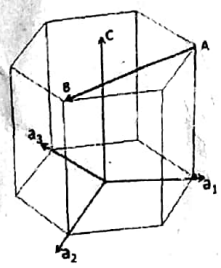
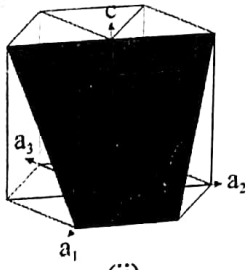
[9]

s) Determine the indices (showing steps) for the following directions and shaded planes:



(i)



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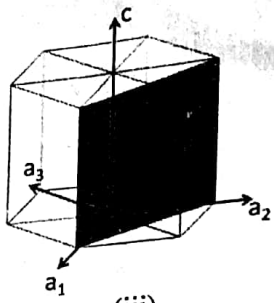

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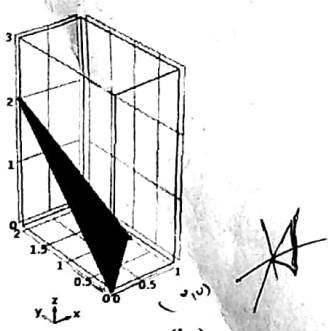

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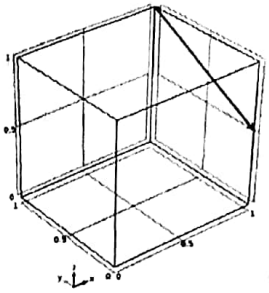

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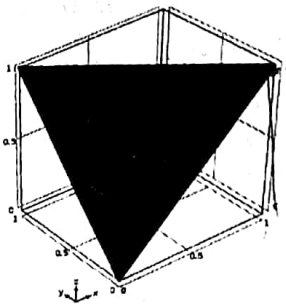

(vi)

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Indian Institute of Technology, Bhubaneswar

School of Electrical Sciences

Introduction to Electronics (EC2L001)

Date: September 24, 2018 **ECE/EE**

Time: 2 hours Autumn Mid-Semester Examination 2018 Maximum Marks: 60

Instructions

1. If you make any assumptions, mention them clearly.
2. Explain your concepts or ideas using a plot or a diagram with minimal explanation wherever possible.
3. Draw the diagrams and plots clearly and label the axes properly.
4. Be precise and answer to the point. "Keep it short, and simple".

Wish you all the best

Q.1) Draw the output voltage v_0 waveform for each diode circuit shown in Figure 1. Assume the input to all the circuit is 10 V (peak) sinusoidal signal. The battery potential V_1 or V_2 is less than 10 V and V_1 is greater than V_2. You may assume that the diode is ideal and the RC time constant is large compared to the frequency of the input signal.

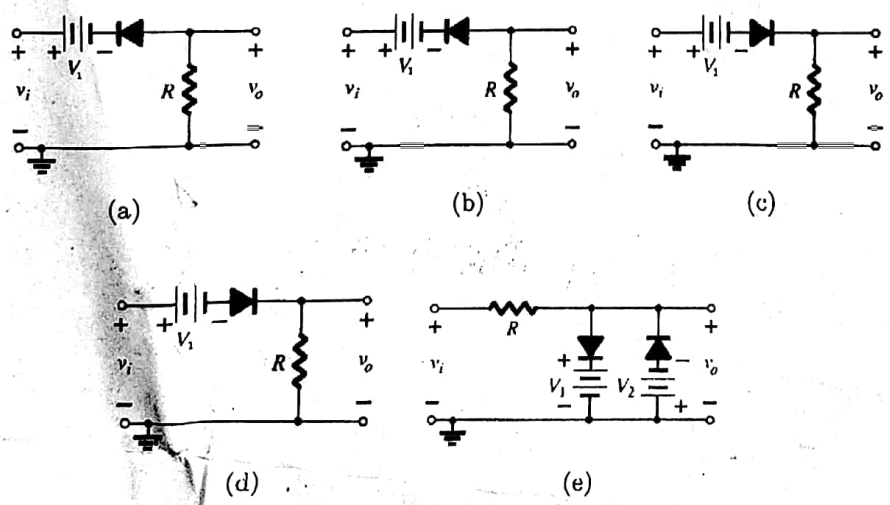

Figure 1: Diode Circuits

[5 Marks]

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Q.2) Given the diode circuit in Figure 2, how do you determine the load line? Explain different steps with relevant diagrams.

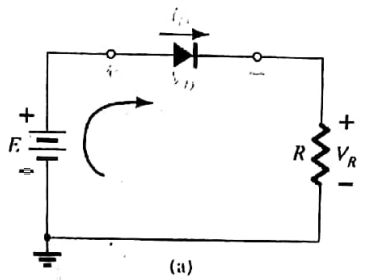

Figure 2: Series Diode Configuration

[5 Marks]

Q.3) Draw the energy band diagram of a *pn* junction diode for the following conditions.

(a) before diffusion
(b) after diffusion
(c) in forward-bias
(d) in reverse-bias

[5 Marks]

Q.4) The unregulated voltage in the Figure 3 may vary from 18 to 25 V, and the load current may vary from 1 to 25 mA. Will the zener regulator stop regulating under these conditions? If so, what is the maximum value for R_S?

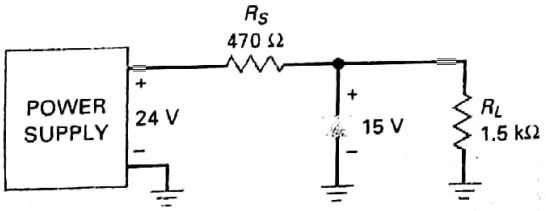

Figure 3: Zener Diode Regulator Circuit

[5 Marks]

Q.5) What is the dc output voltage in Figure 4? The ripple? Sketch the output waveform.

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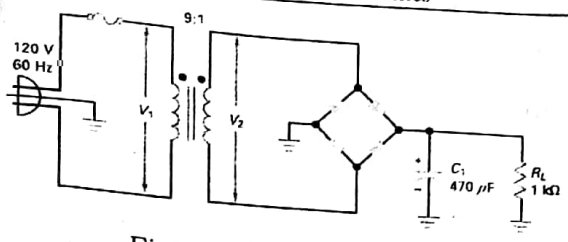

Figure 4: Rectifier Circuit

[5 Marks]

Q.6) Draw the input and output characteristics of both Common Emitter (CE) and Common Base (CB) configuration including the circuit diagrams to characterize them. Answer the following questions with respect to the above transistor configurations and their characteristics:

(a) What is **Early Effect** or **Base Width Modulation**? How does it effect the characteristics of both CB and CE configurations.

(b) Why the slope of CE output characteristics is more compared to the slope of CB output characteristics?

(c) Why the CB output characteristics shows significant amount of current event at $V_{CB} = 0$ compared to the CE characteristics at $V_{CE} = 0$.

[10 Marks]

Q.7) Derive the relationships among α, β, and γ of a BJT transistor. [2 Marks]

Q.8) Compare CE, CB, CC amplifier with respect to voltage gain, current gain, input impedance, and output impedance. [3 Marks]

Q.9) In Figure 5 what is the ac voltage at the first base? At the second base? Across the load resistor?

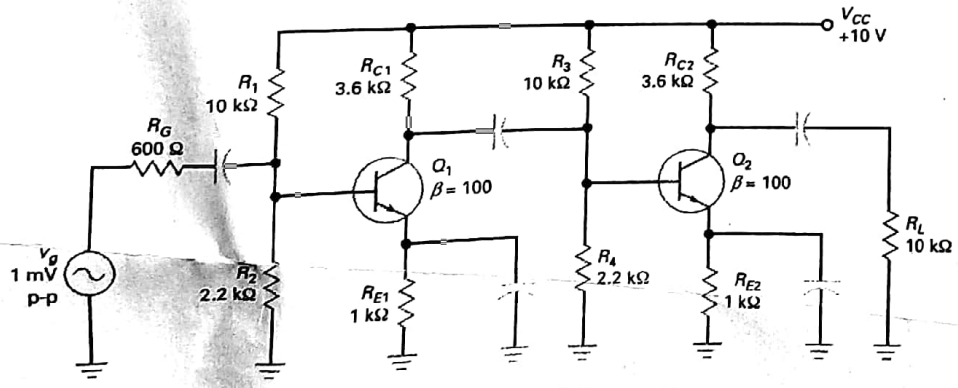

Figure 5: Multi-Stage CE amplifier

$-\frac{V_{2PL}}{V_1}$ $-A_i \times R_L$ / Z_i 1.895×10^{-5} 527.7 0.0455

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[5 Marks]

Q.10) What is the output voltage of Figure 6?

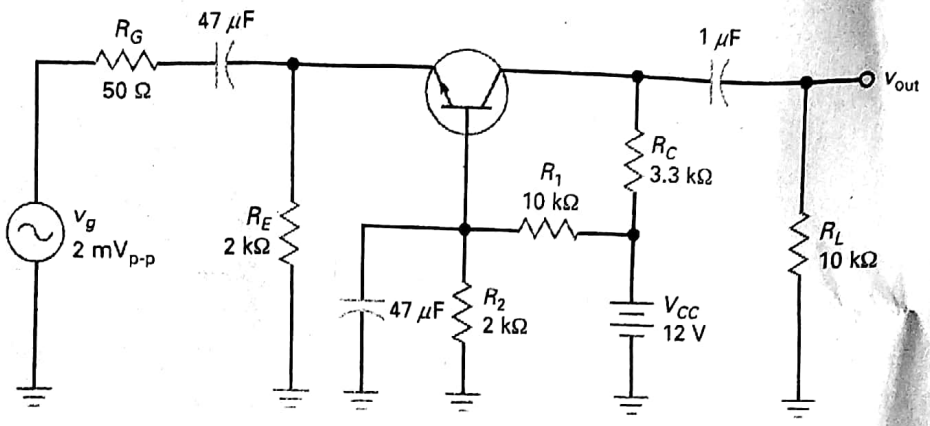

Figure 6: Amplifier Circuit

[5 Marks]

Q.11) (a) Explain the operating principle of a Junction Filed Effect Transistor (JFET).

(b) Draw the output characteristics of a JFET.

(c) Define pinch-off voltage and gate-cutoff voltage and state the relation between them.

(d) Define the maximum drain current I_{DSS}.

[5 Marks]

Q.12) (a) Differentiate a Depletion type MOSFET from a JFET.

(b) Explain the construction and operating principle of both Depletion type and Enhancement type MOSFET and draw the drain curves for both of them.

[5 Marks]

33

32.73

0.861 mV

End of exam

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